

Counting

The Basics of Counting

Basic Counting principles

THE PRODUCT RULE

Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 n_2$ ways to do the procedure.

Example: A new company with just two employees, Sahin and Parvin, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution:

An office can be assigned to Sahin in 12 ways.

After assigning an office to Sahin, an office can be assigned to Parvin in 11 different ways.

Therefore, by the product rule, there are $12 \cdot 11 = 132$ ways to assign offices to these two employees.

Example: The chairs of an auditorium are to be labeled with a letter and a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

Solution:

A letter can be assigned in 26 different ways

An integer can be assigned in 100 different ways

The different ways that a chair can be labeled is $26 \cdot 100 = 2600$

Therefore, the largest number of chairs that can be labeled differently is 2600.

Example: How many different bit strings are there of length seven?

Solution:

Each of the seven bits can be chosen in two ways

Therefore, there are a total of $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^7 = 128$ different bit strings of length seven.

Example: What is the value of k after the following code has been executed?

```
 $k := 0$   
for  $i_1 := 1$  to  $n_1$   
    for  $i_2 := 1$  to  $n_2$   
        .  
        .  
        .  
        for  $i_m := 1$  to  $n_m$   
             $k := k + 1$ 
```

Solution:

The initial value of k is zero

Each time the nested loop is traversed, 1 is added to k

The nested loop is traversed $n_1 n_2 \dots n_m$ times

Therefore, the final value of k is $n_1 n_2 \dots n_m$

THE SUM RULE

If a first task can be done in n_1 ways and second task in n_2 ways, and if these tasks cannot be done at the same time, then there are $n_1 + n_2$ ways to do one of these tasks.

Example: Suppose that either a member of the mathematics faculty or a student who is mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors?

Solution:

Choosing a member of the mathematics faculty can be done in 37 ways.

Choosing mathematics major can be done in 83 ways.

There are $37 + 83 = 120$ possible ways to pick this representative.

Example: A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. How many possible projects are there to choose from?

Solution:

The student can choose a project from the first list in 23 ways, from the second list in 15 ways, and from the third list in 19 ways.

Hence, there are $23 + 15 + 19 = 57$ projects to choose from.

Example: What is the value of k after the following code has been executed?

```
 $k := 0$   
for  $i_1 := 1$  to  $n_1$   
     $k := k + 1$   
for  $i_2 := 1$  to  $n_2$   
     $k := k + 1$   
    .  
    .  
    .  
for  $i_m := 1$  to  $n_m$   
     $k := k + 1$ 
```

Solution:

The initial value of k is zero.

This block of code is made up of m different loops. Each time a loop is traversed, 1 is added to k .

The final value of k is $n_1 + n_2 + \dots + n_m$.

More Complex Counting problems

Example: In a version of the computer language BASIC, the name of a variable is a string of one or two alphanumeric characters, where uppercase and lowercase letters are not distinguished. Moreover, a variable name must begin with a letter and must be different from the five strings of two characters that are reserved for programming use. How many different variables names are there in this version of BASIC?

Solution:

Let V be the number of different variable names in this version of BASIC.

Let V_1 be the number of these that are one character long.

$$V_1 = 26$$

Let V_2 be the number of these that are two characters long

$$V_2 = 26 \cdot 36 - 5 = 931$$

$$V = V_1 + V_2 = 26 + 931 = 957$$

Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution:

Let P be the total number of possible passwords, and P_6 , P_7 , and P_8 denote the number of possible passwords of length 6, 7, and 8, respectively.

The number of strings of six characters is 36^6

The number of strings with no digits is 26^6

$$\therefore P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560$$

Similarly,

$$P_7 = 36^7 - 26^7 = 78,367,164,096 - 8,031,810,176 = 70,332,353,920$$

$$P_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880$$

$$P = P_6 + P_7 + P_8 = 1,867,866,560 + 70,332,353,920 + 2,612,282,842,880 = 2,684,483,063,360$$

The Inclusion-Exclusion Principle

When a part of two tasks can be done at the same time, we cannot use the sum rule to count the number of ways to do one of the two tasks. Adding the number of ways to do each task leads to an overcount, since the ways to do the part in both tasks are counted twice. To correctly count the number of ways to do one of the two tasks, we add the number of ways to do each of the two tasks and then subtract the number of ways to do the part of both tasks. This technique is called the **principle of inclusion-exclusion**.

Example: How many bit strings of length eight start with a 1 bit or end with the two bits 00?

Solution:

Constructing a bit string of length eight beginning with a 1 bit can be done in $2^7 = 128$ ways.

Constructing a bit string of length eight ending with the two bits 00 can be done in $2^6 = 64$ ways.

Constructing a bit string of length eight that begins with a 1 and ends with 00 can be done in $2^5 = 32$ ways.

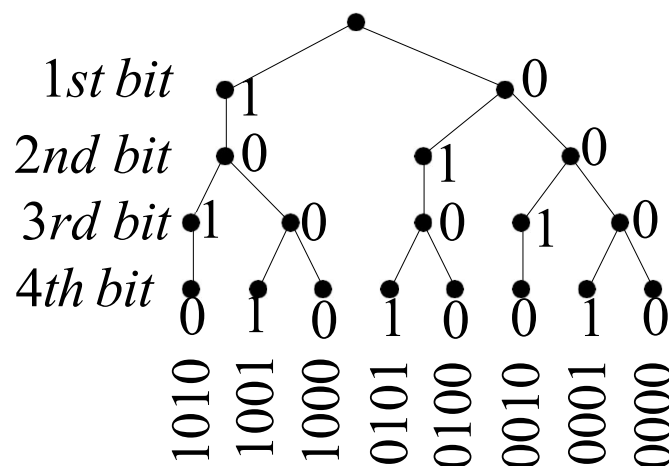
Consequently, the number of bit strings of length eight that begin with a 1 or end with a 00 equals $128 + 64 - 32 = 160$.

Tree Diagrams

Counting problems can be solved using **tree diagrams**. To use trees in counting, we use a branch to represent each possible choice. We represent the possible outcomes by the leaves.

Example: How many bit string of length four do not have two consecutive 1s?

Solution:



The tree diagram displays all bit strings of length four without two consecutive 1s. We see that there are eight bit strings of length four without two consecutive 1s.

The Pigeonhole Principle

Theorem 1: THE PIGEONHOLE PRINCIPLE

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is **at least one box** containing two or more (**at least two**) of the objects.

Proof.

Suppose that none of the k boxes contains more than one object. Then the total number of objects would be at most k . This is a contradiction, since there are at least $k + 1$ objects.

Example: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example: In any group of 27 English words, there must be at least two that begin with the same letter, since there are 26 letters in the English alphabet.

Example: How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100?

Solution:

There are 101 possible scores on the final. The pigeonhole principle shows that among any $101 + 1 = 102$ students there must be at least 2 students with the same score.

The Generalized Pigeonhole principle

Theorem 2: THE GENERALIZED PIGEONHOLE PRINCIPLE

If N objects are placed into k boxes, then there is at least one box containing at least $r = \lceil N/k \rceil$ objects.

Proof:

Suppose that none of the boxes contain more than $\lceil N/k \rceil - 1$ objects. Then, the total number of objects is at most

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N$$

This is a contradiction, since there are a total of N objects.

A common type of problem asks for the minimum number of objects N so that at least r of these objects must be in one of k boxes when these objects are distributed among the boxes.

When we have N objects, the generalized pigeonhole principle tells us that there must be at least r objects in one of the boxes as long as

$$\lceil N/k \rceil \geq r$$

$$N/k + 1 > r$$

$$N/k > (r - 1)$$

$$N > k(r - 1)$$

$N = k(r - 1) + 1$ is the smallest integer satisfying the condition

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

Example: What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

Solution:

Let

N = minimum number of students to satisfy the condition = ?

k = total number of grades = 5

r = minimum number of students receiving the same grade = 6

$$N = k(r - 1) + 1 = 5(6 - 1) + 1 = 26$$

Permutations

EXAMPLE 1 In how many ways can we select three students from a group of five students to stand in line for a picture? In how many ways can we arrange all five of these students in a line for a picture?

Solution: First, note that the order in which we select the students matters. There are five ways to select the first student to stand at the start of the line. Once this student has been selected, there are four ways to select the second student in the line. After the first and second students have been selected, there are three ways to select the third student in the line. By the product rule, there are $5 \cdot 4 \cdot 3 = 60$ ways to select three students from a group of five students to stand in line for a picture.

To arrange all five students in a line for a picture, we select the first student in five ways, the second in four ways, the third in three ways, the fourth in two ways, and the fifth in one way. Consequently, there are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways to arrange all five students in a line for a picture.

EXAMPLE 2 How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution: Because it matters which person wins which prize, the number of ways to pick the three prize winners is the number of ordered selections of three elements from a set of 100 elements, that is, the number of 3-permutations of a set of 100 elements. Consequently, the answer is

$$P(100, 3) = 100 \cdot 99 \cdot 98 = 970,200.$$

EXAMPLE 3 Suppose that there are eight runners in a race. The winner receives a gold medal, the second place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?

Solution: The number of different ways to award the medals is the number of 3-permutations of a set with eight elements.

Hence, there are $P(8, 3) = 8 \cdot 7 \cdot 6 = 336$ possible ways to award the medals.

EXAMPLE 4 Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution: The number of possible paths between the cities is the number of permutations of seven elements, because the first city is determined, but the remaining seven can be ordered arbitrarily. Consequently, there are $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$ ways for the saleswoman to choose her tour. If, for instance, the saleswoman wishes to find the path between the cities with minimum distance, and she computes the total distance for each possible path, she must consider a total of 5040 paths!

EXAMPLE 5 How many permutations of the letters $ABCDEFGH$ contain the string ABC ?

Solution: Because the letters ABC must occur as a block, we can find the answer by finding the number of permutations of six objects, namely, the block ABC and the individual letters D , E , F , G , and H . Because these six objects can occur in any order, there are $6! = 720$ permutations of the letters $ABCDEFGH$ in which ABC occurs as a block.

Combinations

EXAMPLE 1 How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?

Solution: The answer is given by the number of 5-combinations of a set with 10 elements. The number of such combinations is

$$C(10, 5) = 10!/(5! \cdot 5!) = 252.$$

EXAMPLE 2 A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission (assuming that all crew members have the same job)?

Solution: The number of ways to select a crew of six from the pool of 30 people is the number of 6-combinations of a set with 30 elements, because the order in which these people are chosen does not matter. The number of such combinations is

$$C(30, 6) = 30!/(6! \cdot 24!) = 593,775.$$